Nepal PreTST 2026 Solutions

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§1 Problems

Problem 1.1. Let $a_1, a_2, \ldots, a_{2n+2}$ be positive real numbers. Prove that they can be permuted into numbers $u_0, v_0, u_1, v_1, \ldots, u_n, v_n$ such that the polynomial

$$P(x) = u_0 - v_0 x + u_1 x^2 - v_1 x^3 + \cdots + u_n x^{2n} - v_n x^{2n+1}$$

has at least one real root, and every real root of P is greater than or equal to 1.

Problem 1.2. Find all functions $f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ such that for all $m \in \mathbb{N}$,

$$f(m-1) \mid m \qquad \text{and} \qquad m+1 \mid T_{2026}(m),$$

where $T_1(m) = f(m)$ and $T_{k+1}(m) = f\big(m + T_k(m)\big)$ for all $k \geq 1$.

Problem 1.3. Let ABC be an acute scalene triangle with $\angle BAC = 60^\circ$. Denote by O and H the circumcenter and orthocenter of $\triangle ABC$ respectively. Let M be the midpoint of arc BC on (ABC) not containing A, and let ray $\overrightarrow{MH}$ intersect (ABC) a second time at P. Show that the lines AP, OH and BC pass through a common point.

Problem 1.4. Consider the standard triangular grid of unit equilateral triangles forming a larger equilateral triangle of side length n. Initially, only the top unit triangle is colored black, and all other unit triangles are colored white.

A **move** consists of choosing a black unit triangle and simultaneously flipping the colors of **all** unit triangles that share an edge with it; that is, each such neighboring triangle changes from white to black or from black to white. Determine all positive integers n for which it is possible, after a sequence of moves, for all unit triangles in the grid to be black at the same time.

Problem 1.5. Let $\sigma(n)$ denote the divisors sum of n, and let $d(n)$ denote the number of divisors of n. Prove that for any nonnegative integer k, the equation $\sigma(n) = d(n)^k$ has finitely many solutions.

Problem 1.6. Let ω_1 and ω_2 be two circles intersecting at distinct points X and Y. Let ℓ be their common tangent closer to Y, intersecting ω_1 at A and ω_2 at B. The tangent to ω_1 at Y meets ω_2 again at C with $C \neq Y$. Let the line XB meet ω_1 again at K with $K \neq X$. Let $D = BC \cap AY$, and let $E = YK \cap CX$.

Prove that D, Y, X, E are concyclic. (Four points are concyclic if there exists a circle containing all four points.)

Problem 1.7. For each positive integer n, let $\kappa(n)$ denote the unique squarefree positive integer such that $\frac{n}{\kappa(n)}$ is a perfect square. Let G be a connected simple graph whose edges are labeled by positive integers w_e. Assume that for every cycle C in G, the product $\prod_{e\in C} w_e$ is a perfect square. Prove that there exist squarefree positive integers s_v assigned to the vertices $v \in V(G)$ such that for every edge $uv \in E(G)$ there exists a positive integer t_{uv} with $w_{uv} = \kappa(s_u s_v)\, t_{uv}^2$.

Problem 1.8. For all positive reals a, b, c, prove the inequality

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \ \geq \ \frac{3}{2} + \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2(a+b+c)^2}.$$

Determine when equality holds.

§2 Solutions

§2.1 Problem 1, proposed by Prajit Adhikari(Nepal)

Problem 2.1. Let $a_1, a_2, \ldots, a_{2n+2}$ be positive real numbers. Prove that they can be permuted into numbers $u_0, v_0, u_1, v_1, \ldots, u_n, v_n$ such that the polynomial

$$P(x) = u_0 - v_0 x + u_1 x^2 - v_1 x^3 + \cdots + u_n x^{2n} - v_n x^{2n+1}$$

has at least one real root, and every real root of P is greater than or equal to 1.

Proof. Obviously P has one real root due to it having an odd degree. We then group the numbers into $n+1$ pairs like (u_0, v_0), (u_1, v_1), $\cdots (u_n, v_n)$ such that for each $0 \le k \le n$, $u_k \ge v_k$. We now observe the behavior of the following expression:

$$u_k x^{2k} - v_k x^{2k+1} = x^{2k}((u_k - v_k) + v_k(1-x))$$

Which, for $x < 1$, is nonnegative. Moreover, consider the case when $x = 0$, we get for $k = 0$ that $u_0 - v_0(0) = u_0 > 0$ so at least one of these pairs is strictly positive for all $x < 1$ and we conclude that

$$P(x) > 0 \text{ for all } x < 1.$$

FTSOC, if $r < 1$ was a real root of P, we would have $P(r) = 0$ and $P(r) > 0$, contradiction. Therefore every real root r of P satisfies $r \ge 1$. $\square$

§2.2 Problem 2, proposed by Shining Sun(USA)

Problem 2.2. Find all functions $f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ such that for all $m \in \mathbb{N}$,

$$f(m-1) \mid m \qquad \text{and} \qquad m+1 \mid T_{2026}(m),$$

where $T_1(m) = f(m)$ and $T_{k+1}(m) = f\big(m + T_k(m)\big)$ for all $k \geq 1$.

Proof. We proceed by downward induction. Assume, for some $k \leq 2026$, that $m+1 \mid T_k(m)$. Then,

$$m+1 \mid T_k(m) = f(m + T_{k-1}(m)) \mid m + 1 + T_{k-1}(m)$$

Therefore we get that $m+1 \mid T_k(m) \implies m+1 \mid T_{k-1}(m)$, so we reduce this to $m+1 \mid f(m)$. Combining with the first condition we get that $f(m) \mid m+1$ which gives $f(m) = m+1$ for $m > 0$. However setting $m = 1$ in the first condition gives us $f(0) = 1$, therefore $f(m) = m+1$ for all $m \geq 0$. $\square$

§2.3 Problem 3, proposed by Lasitha Jayasinghe(Sri Lanka)

Problem 2.3. Let ABC be an acute scalene triangle with $\angle BAC = 60^\circ$. Denote by O and H the circumcenter and orthocenter of $\triangle ABC$ respectively. Let M be the midpoint of arc BC on (ABC) not containing A, and let ray $\overrightarrow{MH}$ intersect (ABC) a second time at P. Show that the lines AP, OH and BC pass through a common point.

Proof. It suffices to show that quadrilaterals $BOHC$ and $APHO$ are cyclic, from which the radical axis theorem finishes.

Claim 2.4 — Quadrilateral $BOHC$ is cyclic.

Proof. We have

$$\angle BHC = 180^\circ - \angle BAC = 120^\circ = 2\angle BAC = \angle BOC.$$

□

Claim 2.5 — Quadrilateral $APHO$ is cyclic.

Proof. We have that:

$$\begin{aligned}\angle HOA &= 360^\circ - \angle AOB - \angle BOH \\ &= 180^\circ - 2\angle ACB + \angle HCB \\ &= 270^\circ - 2\angle ACB - \angle CBA\end{aligned}$$

Also, we have

$$\begin{aligned}\angle APH &= 180^\circ - \angle MBA \\ &= 180^\circ - (\angle MBC + \angle CBA) \\ &= 180^\circ - (90^\circ - \frac{\angle BOC}{2} + \angle CBA) \\ &= 180^\circ - (90^\circ - \angle BAC + \angle CBA) \\ &= 150^\circ - \angle CBA\end{aligned}$$

Therefore we get that $\angle HOA + \angle APH = 420^\circ - 2(\angle ACB + \angle CBA) = 180^\circ$. □

Therefore radical axis on (ABC), $APHO$, and $BOHC$ completes. However, it doesn't. We are left to prove that the three lines actually concur at some point, and are not parallel.

Claim 2.6 — The lines AP, OH, and BC are not parallel.

Proof. Assume not, for the sake of contradiction. Then, $BHOC$ becomes a cyclic trapezoid so $BH = OC$. Some simple length chasing gives

$$BH = 2CO\cos(\angle CBA)$$

Which forces $\angle CBA = 60^\circ$, contradicting $\triangle ABC$ scalene. Therefore, the three lines are not parallel. □

And we're actually done. □

§2.4 Problem 4, proposed by Shining Sun(USA)

Problem 2.7. Consider the standard triangular grid of unit equilateral triangles forming a larger equilateral triangle of side length n. Initially, only the top unit triangle is colored black, and all other unit triangles are colored white.

A **move** consists of choosing a black unit triangle and simultaneously flipping the colors of **all** unit triangles that share an edge with it; that is, each such neighboring triangle changes from white to black or from black to white. Determine all positive integers n for which it is possible, after a sequence of moves, for all unit triangles in the grid to be black at the same time.

Proof. □

§2.5 Problem 5, proposed by Basu Dev Karki(Nepal)

Problem 2.8. Let $\sigma(n)$ denote the divisors sum of n, and let $d(n)$ denote the number of divisors of n. Prove that for any nonnegative integer k, the equation $\sigma(n) = d(n)^k$ has finitely many solutions.

Proof. □

§2.6 Problem 6, proposed by Kritesh Dhakal(Nepal)

Problem 2.9. Let ω_1 and ω_2 be two circles intersecting at distinct points X and Y. Let ℓ be their common tangent closer to Y, intersecting ω_1 at A and ω_2 at B. The tangent to ω_1 at Y meets ω_2 again at C with $C \neq Y$. Let the line XB meet ω_1 again at K with $K \neq X$. Let $D = BC \cap AY$, and let $E = YK \cap CX$.

Prove that D, Y, X, E are concyclic. (Four points are concyclic if there exists a circle containing all four points.)

Proof. □

§2.7 Problem 7, proposed by Prajit Adhikari(Nepal), Shreyash Sharma Bastola(Nepal)

Problem 2.10. For each positive integer n, let $\kappa(n)$ denote the unique squarefree positive integer such that $\frac{n}{\kappa(n)}$ is a perfect square. Let G be a connected simple graph whose edges are labeled by positive integers w_e. Assume that for every cycle C in G, the product $\prod_{e\in C} w_e$ is a perfect square. Prove that there exist squarefree positive integers s_v assigned to the vertices $v \in V(G)$ such that for every edge $uv \in E(G)$ there exists a positive integer t_{uv} with $w_{uv} = \kappa(s_u s_v)\, t_{uv}^2$.

Proof. □

§2.8 Problem 8, proposed by Prajit Adhikari(Nepal), Kritesh Dhakal(Nepal)

Problem 2.11. For all positive reals a, b, c, prove the inequality

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \;\geq\; \frac{3}{2} + \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2(a+b+c)^2}.$$

Determine when equality holds.

Proof. □